2 parâmetros

$$y = f(\theta) = f(\bar{\theta}) + \sum_{m=1}^{\infty}\left(\frac{1}{m!}\sum_{i=1}^{N}\left(\frac{d f(\bar{\theta})}{d\theta_i}(\theta_i - \bar{\theta}_i)^m\right)\right)$$

$$y = f(\bar{\theta}) + \frac{d f(\bar{\theta})}{d\theta_1}(\theta_1 - \bar{\theta}_1) + \frac{d f(\bar{\theta})}{d\theta_2}(\theta_2 - \bar{\theta}_2)$$

$$y - f(\bar{\theta}) = \frac{d f(\bar{\theta})}{d\theta_1}\Delta\theta_1 + \frac{d f(\bar{\theta})}{d\theta_2}\Delta\theta_2$$

$\Delta y = \rightarrow$

$$\Delta\theta = (J^T J)^{-1} J^T \Delta y$$

$$\bar{\theta}_{novo} = \bar{\theta} + \Delta\theta$$

x^{θ_1}

$x = [0,1,2]^T$

$y = [3,2,1]^T$

$y = \theta_1 e^{\theta_2 x}$

$y = f(x,\theta)$

$$y = \frac{d f(\bar{\theta})}{d\theta_1}\Delta\theta_1 + \frac{d f(\bar{\theta})}{d\theta_2}\Delta\theta_2$$

$$\frac{d f(\bar{\theta})}{d\theta_1} = e^{\bar{\theta}_2 x}$$

$$\frac{d f(\bar{\theta})}{d\theta_2} = x\bar{\theta}_1 e^{\bar{\theta}_2 x}$$

$$y - \bar{\theta}e^{\bar{\theta}_2 x} = \left(e^{\bar{\theta}_2 x}\right)\Delta\theta_1 + \left(x\bar{\theta}e^{\bar{\theta}_2 x}\right)\Delta\theta_2$$

$$\begin{bmatrix}\Delta y_1\\ \Delta y_2\\ \Delta y_3\end{bmatrix} = \begin{bmatrix} e^{\bar{\theta}_2 x_1} & x_1\bar{\theta}_1 e^{\bar{\theta}_2 x_1}\\ e^{\bar{\theta}_2 x_2} & x_2\bar{\theta}_1 e^{\bar{\theta}_2 x_2}\\ e^{\bar{\theta}_2 x_3} & x_3\bar{\theta}_1 e^{\bar{\theta}_2 x_3}\end{bmatrix}\cdot\begin{bmatrix}\Delta\theta_1\\ \Delta\theta_2\end{bmatrix}$$

$\Delta y = y - \bar{\theta}_1 e^{\bar{\theta}_2}$

$$\frac{d f\bar{\theta}}{d\theta_1} = \ln(x)\cdot x^{\theta}$$

$$f(\theta_1) = \theta_1 + \theta_2 e^{\theta_3 x} + \theta_4 e^{\theta_5 x} + \theta_6 e^{\theta_7 x}$$

$$\frac{d f\bar{\theta}}{d\theta_1} = 1$$

$$\frac{d f\bar{\theta}}{d\theta_2} = e^{\theta_3 x}$$

p/ i > n

$$J = \left[\int e^{\theta_{i+1}x} \times\theta_i e^{\theta_{i+1}x}\right]$$

$$\frac{d f\bar{\theta}}{d\theta_3} = x\theta_2 e^{\theta_3 x}$$

$$\frac{d f\bar{\theta}}{d\theta_4} = e^{\theta_5 x}$$

$$\frac{d f\bar{\theta}}{d\theta_5} = x\theta_4 e^{\theta_5 x}$$

grau $n \rightarrow 3\cdot(n-1)+1$ ← número de thetas

$$f(\theta_1) = \theta_1 + \theta_2\tanh(\theta_3 x + \theta_4) + \theta_5\tanh(\theta_6 x + \theta_7)\Big\}\text{ grau 3}$$

th(1) → constante

de $i = 1 : n-2$

$$y_n = \left[y_n\ \theta_{3i-2}\tanh(\theta_{3i}x + \theta_{3i+2})\right]$$

fim

$2 \rightarrow 5 \rightarrow 8 \rightarrow 11 \rightarrow 14$
$3 \rightarrow 6 \rightarrow 9 \rightarrow 12 \rightarrow 15$
$4 \rightarrow 7 \rightarrow 10 \rightarrow 13 \rightarrow 16$

$$\frac{d\bar{\theta}}{d\theta_1} = 1$$

$$\frac{d\bar{\theta}}{d\theta_2} = \tanh(\theta_3 x + \theta_4)$$

$$\frac{d\bar{\theta}}{d\theta_3} = \theta_2\cdot x\cdot\sec^2(\theta_3 x + \theta_4)$$

$$\frac{d\bar{\theta}}{d\theta_4} = \theta_2\cdot\sec^2(\theta_3 x + \theta_4)$$

fim

Prova 2018.1

Q.1)

4 controles

$$\theta_1 + \theta_2\tanh(\theta_3 x_1 + \theta_4 x_2 + \theta_5 x_3 + \theta_6 x_4 + \theta_7)$$

(6-4) (6-3) (6-2) (6-1) (6) (6+1)

num de parâmetros $= (6\cdot n) + 2 - 1$

$d_2 = 2$

Q.2)

$$1 + (0.1 + 0.26j)e^{(-3+5(6)j)t} + (0.1 - 0.26j)e^{(-3-5(6)j)t} - 1.2\,3\,e^{(-3)t}$$

$$(0.1+0.26j)e^{-3}(\cos(6t) + j\sin(6t)) + (0.1-0.26j)e^{-3}(\cos(6t) - j\sin(6t))$$

$$e^{-3t}\left((0.1+0.26j)(\cos(6t)+j\sin(6t)) + (0.1-0.26j)(\cos(6t)-j\sin(6t))\right)$$

$$e^{-3t}(2\cdot 0.1\cos(6t) - 2\cdot 0.26\sin(6t))$$

$$2\cdot e^{-3}(0.1\cos(6t) - 0.26\sin(6t))$$

$$1 + 2\cdot e^{-3t}(0.1\cos(6t) - 0.26\sin(6t)) - 1.2\cdot e^{(-3)t}$$

$$F(s) = \frac{1}{s} + 0.2\frac{s+3}{(s+3)^2+6^2} - 0.5\frac{6}{(s+3)^2+6^2} - \frac{1.2}{s+3}$$

$$Y(s) = \frac{1}{s} + \frac{0.2s - 2.4}{(s+3)^2+6^2} - \frac{1.2}{s+3}$$

$$H(s) = \frac{Y(s)}{U(s)} = \frac{1}{1} + \frac{s(0.2s-2.4)}{(s+3)^2+36} - \frac{1.2s}{s+3} \Rightarrow 1 + \frac{[s+3]\cdot[s(0.2s-2.4)] - 1.2s\cdot[(s+3)^2+36]}{[(s+3)^2+36]\cdot[s+3]} = \frac{(s+3)^3+36(s+3)}{[(s+3)^2+36]\cdot[s+3]} + \frac{[s+3]\cdot[0.2s^2-2.4s] - 1.2s\cdot[(s+3)^2+36]}{[(s+3)^2+36]\cdot[s+3]}$$

$$\theta_2 + \theta_{s-1}\tanh\left(\theta_3 x_2 + \theta_4 x_{s-2} + \theta_5 x_{s-1} + \theta_6\right)$$

Prova (2019)

q1)

$$A(s) = \frac{k_A\left(\frac{1}{\tau_A}\right)}{s\left(s+\frac{1}{\tau_A}\right)} = \frac{A}{s} + \frac{B}{\left(s+\frac{1}{\tau_A}\right)} \Rightarrow \left(s+\frac{1}{\tau_A}\right)A + Bs = k_A\left(\frac{1}{\tau_A}\right)$$

p/ $s = -\frac{1}{\tau_A}$: $B\left(-\frac{1}{\tau_A}\right) = k_A\left(\frac{1}{\tau_A}\right) \Rightarrow B = -k_A$

p/ $s = 0$: $\frac{A}{\frac{1}{\tau_A}} = k\left(\frac{1}{\tau_A}\right) = A = k_A$

$$\frac{k_A}{s} - \frac{k_A}{\left(s+\frac{1}{\tau_A}\right)} = k_A - k_A \cdot e^{(-0.7s)}$$

$A_2 = 10 - 10\cdot e^{-0.7s} \Rightarrow 10\left(1 - e^{-0.7s}\right)$ = resposta ao degrau

$B_2 \Rightarrow 8 - 8\cdot e^{-2.6}$

$$1 - 1e^{-3.5t}$$

$$1 - e^{-3.5t}$$

$$H(s)\cdot U(s) = \frac{1}{s} - \frac{1}{s+3.5}$$

dudas 1) $H(s) = 1 - \frac{s}{s+3.5} = H(s) = \frac{s-(s+3.5)}{s+3.5} = \frac{Y(s)}{U(s)} = \frac{-\theta_2\theta_a}{(s-\theta_a)} = \frac{-3.5}{s+3.5} = 3{,}5\cdot e^{3.5t} = \frac{3.5}{s+3.5}$

dudas 2) $Y(s) = 3 - 3\ e^{-1.75}$

$3 + 0.5e^{-1.2} - 3.5e^{-1.85} \Rightarrow H(s) = 3 + \frac{0.5s}{s+1.2} - \frac{3.5s}{s+1.85}$

$Y(s) = \frac{3}{s} - \frac{33}{s+1.75}$

$H(s) = 3 - \frac{3.25s}{s+1.75} \rightarrow \frac{3(s+1.75) - 3.25s}{s+1.75} = \frac{-0.25s + 5.25}{s+1.75} =$

$3 + \frac{0.5s(s+1.85) - 3.5s(s+1.2)}{(s+1.2)(s+1.85)}$

$\frac{3 - 1.2s - 1.85}{(s+1.2)(s+1.85)} = \frac{20.5}{(s+0.7)(s+1.85)}$

$3 + \frac{0.5s^2 + 0.92s - 3.5s^2 - 7.2s}{s^2 + 1.85s + 1.2s + 2.22.00}$

dudas 3) $Y(s) = 2 - (1+0.5j)e^{(-0.5+0.85j)} - (1-0.5j)e^{(-0.5-0.85j)}$

$$2 - \frac{1+0.5j}{s+0.5-0.85j} - \frac{1-0.5j}{s+0.5+0.85j}$$

$$\frac{2\cdot(-0.5+0.85j)\cdot(-0.5-0.85j)}{(s+0.5-0.85j)\cdot(s+0.5+0.85j)} = \frac{(-1+1.7j)(-0.5-0.85j)}{(s-0.5)^2 + (-0.85)^2} = +0.5 - 0.85j + 0.85j + 1.445$$

a $\quad$ b

$s^2 + s(a+b) + ab$

$$= \frac{1.945}{s^2 + 2 + 0.9725}$$